

Blockchain-Based Tokenization and Management of Land Documents: A Decentralized Approach for Secure Ownership and Transfer

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Abstract—Blockchain technology is transforming real estate with its transparency and security. Its decentralized, immutable ledger ensures all transaction records are tamper-proof and visible. This paper addresses a critical flaw in the traditional market: fragmented systems that often overlook approvals from inspectors and lenders. By leveraging the Ethereum blockchain and its smart contracts, Block realty automates agreements, eliminating intermediaries and enhancing efficiency. The paper utilizes the ERC-721 standard to tokenize real estate assets, converting properties into unique digital tokens. This fundamentally changes asset management, enhancing market liquidity and global accessibility. A key feature is a comprehensive three-step approval process, ensuring all stakeholders-buyers, lenders, inspectors, and sellers are actively engaged. Each party's approval is immutably recorded on the blockchain, creating a verifiable chain of custody that boosts transaction integrity. The use of Non-Fungible Tokens (NFTs) also facilitates fractional ownership, allowing multiple investors to jointly own a single property. This democratizes real estate investment by lowering the barrier to entry. For data management and cost optimization, Blockrealty integrates the Inter Planetary File System (IPFS), a decentralized storage protocol for securely housing property images and metadata. Additionally, seamless Metamask integration provides a user-friendly interface. Looking ahead, upcoming Ethereum advancements like the Pectra upgrade are poised to enhance Ethereum's scalability and efficiency, further strengthening the foundation for platforms like Blockrealty by enabling them to handle transactions more effectively and reliably.

Index Terms—Blockchain, Tokenization, Real Estate, Smart Contracts, NFTs, IPFS

I. INTRODUCTION

Blockchain technology and Non-Fungible Tokens (NFTs) are transforming the real estate investment landscape by converting physical assets into blockchain-recorded digital tokens, where NFTs represent both ownership and intrinsic value. This innovation introduces fractional ownership, enhances liquidity and flexibility, and significantly reduces entry barriers for smaller investors. The inherent transparency of blockchain ensures verifiable ownership records and tamper-proof transaction histories, thereby minimizing fraud and

disputes. Tokenization further reduces conventional transaction costs and enables global trading of assets, allowing investors to diversify portfolios across residential, commercial, and specialized properties. Smart contracts automate lease agreements, payments, and title transfers, streamlining management and improving operational efficiency.

Integrating ERC-721 tokens into real estate is pivotal in modernizing property ownership, as these unique tokens on the Ethereum blockchain securely represent assets as transparent digital entities. A key feature of ERC-721 is fractional ownership, allowing high-value properties to be divided into tradable digital shares, improving liquidity and democratizing participation. The global traceability of these tokens across decentralized exchanges expands investment opportunities worldwide. Furthermore, the InterPlanetary File System (IPFS) strengthens this framework by providing decentralized, content-addressable storage for NFT metadata-images, documents, and videos ensuring resilience, authenticity, and long-term accessibility. Despite challenges in data preservation and implementation complexity, ongoing developments underscore blockchain's growing integration and transformative potential within real estate ecosystems.

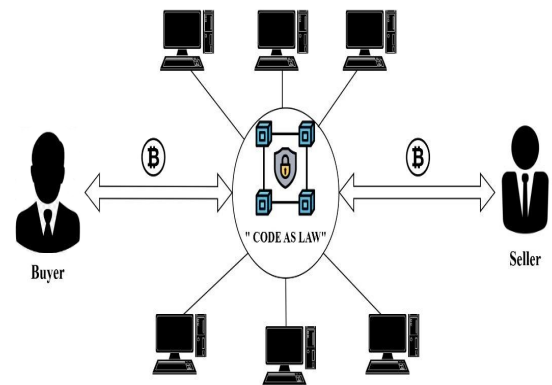


Fig. 1. Smart Contract-Based Real Estate Transaction on Blockchain

II. LITERATURE SURVEY

Property represents one of the most valuable assets in modern economies, and the integration of blockchain technology can significantly enhance efficiency, transparency, and security in real estate operations. As demonstrated by Suryawanshi *et al.* (2023), blockchain enables property sales and transfers through automated smart contracts, ensuring that ownership transactions occur safely and verifiably without intermediaries. Their work illustrates how private blockchain networks can facilitate property registration, while smart contracts automatically execute sales once predefined conditions are met, thus providing immutable proof of ownership transfer [1].

Building on this foundation, Al-haimi *et al.* (2025) conducted a systematic literature review on digital transformation in the real estate industry, covering artificial intelligence (AI), the Internet of Things (IoT), blockchain, augmented reality (AR), virtual reality (VR), and digital twins. Their study highlights how these technologies transform property management, investment evaluation, and consumer interaction. The findings show improvements in operational efficiency, data-driven decision-making, and user engagement. However, the authors also identify challenges such as high implementation costs, cybersecurity risks, and integration difficulties, which require modular technological frameworks and strong security policies for successful adoption [2].

Within this evolving digital landscape, blockchain adoption has become particularly influential in streamlining property registration and verification processes. Research by Pande *et al.* (2022) shows that combining blockchain with smart contracts automates real estate transactions, reducing dependency on intermediaries and lowering operational costs [3]. Silvia (2024) emphasizes the legal and regulatory aspects of tokenization and concludes that blockchain provides a transparent and tamper-proof system that minimizes fraudulent activities and enhances trust in property dealings [4].

Rath *et al.* (2023) expanded on this by developing a Solidity-based smart contract framework that enables fractional ownership of rental assets. Statistical analysis on synthetic data revealed a positive relationship between rental yields and risk–return metrics, confirming that blockchain technology enhances liquidity, reduces costs, and supports broader investment participation [5].

Dwivedi *et al.* (2023) introduced the concept of tokenizing Transferable Development Rights (TDRs) in collaboration with development authorities. Their approach ensures accurate record-keeping, stakeholder verification, and equitable valuation based on public circle rates. This model enables transparent exchanges of development potential between parties and improves accountability in urban land management [6].

A practical implementation of these ideas is seen in the Bitland Project, which leverages blockchain and smart contracts to facilitate fractional ownership, automate commercial real estate investments, and create a decentralized registry for property transactions. The project also identifies several

technical and legal obstacles to blockchain adoption that must be addressed to achieve a global digital property network [7].

However, most existing blockchain models still focus primarily on buyer–seller transactions and neglect important processes such as property inspection and lending. Addressing these gaps requires a more comprehensive ecosystem that combines property verification with blockchain-enabled financial mechanisms.

Existing blockchain models in real estate mainly streamline transactions between buyers and sellers but often neglect essential components like property inspection and lending. The literature survey identifies a gap in incorporating inspectors to verify the authenticity of NFT-based properties and the lack of integrated lending mechanisms to assist buyers during transactions.

III. BLOCKCHAIN IN REAL ESTATE

Blockchain technology continues to transform the real estate industry by providing secure, transparent, and efficient systems for asset management and transfer. According to Shabbir (2021), integrating blockchain with smart contracts helps address persistent issues of transparency, fraud, and transaction delays in real estate operations. It redefines how ownership, verification, and record-keeping are managed within property ecosystems [8].

Ethereum’s blockchain plays a pivotal role in revolutionizing real estate by creating a tamper-proof public ledger for property transactions. This shared ledger eliminates hidden agreements, ensuring that all transaction details are verifiable by all parties. Smart contracts further automate transactions by executing pre-defined conditions, reducing errors, fraud, and the reliance on intermediaries, ultimately increasing trust and efficiency.

The key benefits of blockchain in real estate include:

A. Enhanced Transparency and Security

Blockchain provides a secure and transparent method to record and transfer property ownership, minimizing fraud risks and transactional errors.

B. Improved Liquidity

Tokenization of real estate assets allows fractional ownership, enabling smaller investors to participate in the market a space traditionally dominated by large investors and institutions.

C. Reduced Transaction Costs and Time

By eliminating intermediaries such as brokers and legal representatives, blockchain significantly reduces transaction fees and processing time in property transactions.

D. Cross-Border Transactions

Blockchain simplifies cross-border real estate transactions by standardizing processes, overcoming legal and regulatory challenges across different regions.

E. Automated Compliance and Record Keeping

Smart contracts can automate compliance checks, title verification, and record-keeping, making the entire real estate process more efficient, secure, and error-free.

TABLE I
COMPARISON OF BLOCKCHAIN APPLICATIONS IN REAL ESTATE

Focus	Key Findings	Challenges	Highlighted Benefits
Legal implications of blockchain-based digital tokens in property transfers	Private law can adapt to asset-backed tokenization, enabling a global digital property market	Legal uncertainties across jurisdictions	Global digital market, intermediary-free transactions
Blockchain adoption in real estate	Streamlines property registration and verification, automates transactions with smart contracts	Adoption barriers, regulatory concerns	Lower operational costs, global investment access
Blockchain in real estate	Blockchain enhances transparency and minimizes fraud	Transparency and fraud challenges	Secure, decentralized property transactions
Blockchain platform for fractional ownership	Digital real estate network simplifies investment management	Implementation challenges in the real estate sector	Fractional ownership, streamlined transactions
Tokenization of Transferable Development Rights (TDRs)	Partnership model ensures fair evaluation through circle rates	Land verification, stakeholder identification	Transparent record-keeping, fair compensation
Blockchain in property transfers, tokenization, NFTs	Fractional ownership simplifies transactions and democratizes investments	Slow adoption due to illiquidity	Enhanced transparency, wealth democratization
Tokenization of rental real estate assets	Regression and ANOVA analysis confirm positive rental yield correlation	Risk-return factors, data reliability	Increased liquidity, reduced transaction costs
Tokenization of properties using Ethereum Blockchain	Uses ERC-721 NFTs, IPFS for metadata storage, and <i>Meta-mask</i> for user interaction	Approval process complexity, technical adoption	Enhanced security, transparency, streamlined property investments

With its decentralized and immutable nature, blockchain technology is poised to reshape the global real estate landscape, offering a more accessible, transparent, and efficient market.

IV. CURRENT SCENARIOS AND RESEARCH GAP

A. Existing Landscape

Blockchain technology has significantly improved the efficiency, transparency, and security of real estate transactions. Most current platforms focus on decentralized property marketplaces where verified listings allow direct interactions between buyers and sellers. Smart contracts automate and secure these exchanges by executing predefined conditions on the blockchain, minimizing dependence on brokers and legal intermediaries.

B. Identified Research Gaps

Despite notable progress, several key challenges remain:

1) *Crypto Financing*: Existing systems lack integrated lending mechanisms for buyers with insufficient cryptocurrency. Research on decentralized crypto-financing models, including lender approval, risk evaluation, and repayment through smart contracts, remains limited. Such mechanisms could improve accessibility and accelerate blockchain adoption in real estate.

2) *Property Verification*: Current platforms also lack standardized property inspection and legitimacy checks. Without legal validation or physical inspection, NFT-based assets risk fraudulent listings and ownership disputes. Integrating inspector-driven verification and certification could enhance trust and reduce fraud.

C. Suggested Resolution: Three-Step Approval Process

To address these issues, this study proposes a three-step approval framework involving lenders, inspectors, and sellers

to improve transparency and reliability in blockchain-based property transactions.

1) *Lender Approval*: Smart contracts facilitate crypto-financing by assessing buyer creditworthiness and automating loan approval within the blockchain ecosystem.

2) *Inspector Approval*: Certified inspectors verify property legality and physical condition before tokenization, with reports securely stored on the blockchain for dispute resolution.

3) *Seller Approval*: The seller confirms inspection and payment completion, ensuring all terms are fulfilled before finalizing the contract.

V. METHODOLOGY

The proposed blockchain-based system for real estate transactions leverages Ethereum smart contracts and the InterPlanetary File System (IPFS) to enhance security, transparency, and efficiency. This methodology outlines the key steps involved in the minting, verification, and transaction process of Non-Fungible Tokens (NFTs) representing real estate assets.

Step 1: Data Storage and NFT Creation

The property details, including ownership records, legal documents, and metadata, are securely stored on the IPFS blockchain.

Once uploaded, IPFS returns a unique cryptographic hash that serves as a reference for the stored data.

A Non-Fungible Token (NFT) is then minted on the Ethereum blockchain using the ERC-721 standard. This NFT represents the corresponding real estate asset and contains a link to the IPFS hash for verification purposes.

Step 2: NFT Listing and Bidding Process

The minted NFT is registered for sale through a decentralized marketplace.

Potential buyers can browse available properties and place bids on the listed NFTs.

Smart contracts handle the bidding process, ensuring transparency and automating bid validation.

Step 3: Approval and Verification

The verification and approval process follows three key stages. First, the lender reviews and approves financing requests for buyers lacking sufficient cryptocurrency. Next, a certified property inspector validates the asset's legitimacy, legal compliance, and physical condition to ensure it is free from undisclosed issues. Finally, the seller confirms the transaction terms, authorizing the ownership transfer on the blockchain.

Step 4: Finalizing the Transaction

Upon successful completion of the approval process, the seller accepts the highest valid bid. The smart contract automatically executes the transaction, transferring NFT ownership from the seller to the buyer. The transaction details, including ownership changes, are permanently recorded on the Ethereum blockchain, ensuring immutability and transparency.

Step 5: Post-Transaction Management Buyers can store and manage their NFT-based property assets in their digital wallets (e.g., MetaMask). Future transactions, such as re-sales or property rentals, can be conducted through the same blockchain-based system.

This methodology enables a seamless and trustless mechanism for real estate transactions, minimizing intermediaries while enhancing security and verifiability through blockchain technology. Figure 2 illustrates the process of digital asset exchange using blockchain and non-fungible tokens (NFTs) based on the IPFS network and Ethereum's ERC-721 standard. The seller uploads asset details to IPFS, which generates a unique hash ensuring data integrity. An NFT representing the asset is then minted and listed for sale. A buyer places a bid, and a smart contract governs the transaction, ensuring transparency and automation. Once accepted, ownership is securely transferred to the buyer. This decentralized framework strengthens security, mitigates fraud, and eliminates intermediary dependence in property transactions.

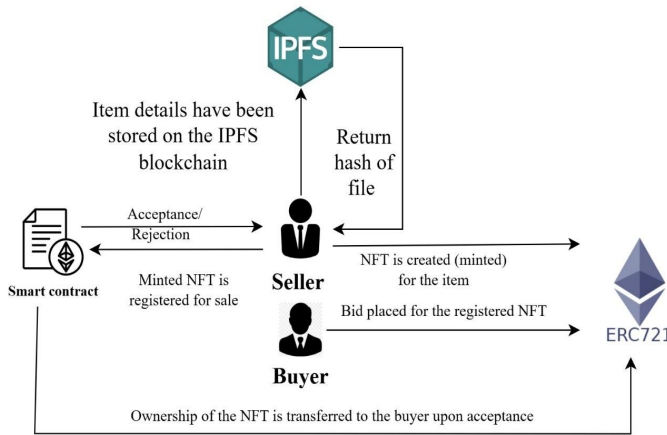


Fig. 2. NFT Marketplace Workflow using IPFS and Ethereum Smart Contracts

This methodology evaluates the performance of a hybrid blockchain system integrating Ethereum, MongoDB, and IPFS. For blockchain performance, it verifies block approval time, Proof of Authority (PoA) configuration, and block production intervals using Geth. Transaction approval times are measured, and PoA is compared with PoW and PoS based on energy use, speed, security, and scalability. Network congestion is tested with 50 simultaneous transactions, analyzing pending queues and gas usage. Gas costs for key operations are documented across Ethereum networks. For MongoDB, query performance is analyzed using indexed vs. non-indexed collections, with optimizations tested on frequently queried fields. MongoDB is benchmarked against relational databases on query speed, schema flexibility, and storage efficiency. Its centralized model is contrasted with IPFS's decentralized storage. For IPFS and CID management, retrieval speeds from local nodes and public gateways are measured. Storage approaches are compared based on speed, latency, redundancy, and security. Data integrity is verified using CID recalculations and tampering detection tests, ensuring content authenticity.

VI. RESULT ANALYSIS

1. Smart Contract Execution & Transaction Efficiency

TABLE II
PERFORMANCE METRICS OF BLOCKCHAIN-BASED REAL ESTATE SYSTEM

Metric	Result	Insight
Smart Contract Execution Time	15–17 sec per transaction	Faster than traditional legal processes
Gas Fees (Ethereum Mainnet)	\$30–\$50 per transaction	High cost limits scalability
Gas Fees (Private Blockchain)	\$5 per transaction	85% cheaper, more sustainable
Tokenization Time	<5 minutes	Quick and transparent ownership transfer
Verification & Approval Time	40% faster than traditional methods	Improved trust and compliance
Network Performance (PoA)	Stable at 50 concurrent transactions	Scalable for real estate applications
MetaMask Integration	100% success rate	User-friendly wallet connectivity

The table above presents the comparative performance of the blockchain-based real estate framework against conventional transaction methods. It demonstrates clear improvements in execution speed, cost efficiency, and scalability achieved through decentralized automation. As illustrated in the figure, the execution of property transactions via Ethereum smart contracts reduced completion time dramatically from nearly 120 seconds in traditional systems to around 17 seconds—highlighting blockchain's capability to streamline and accelerate real estate processes.

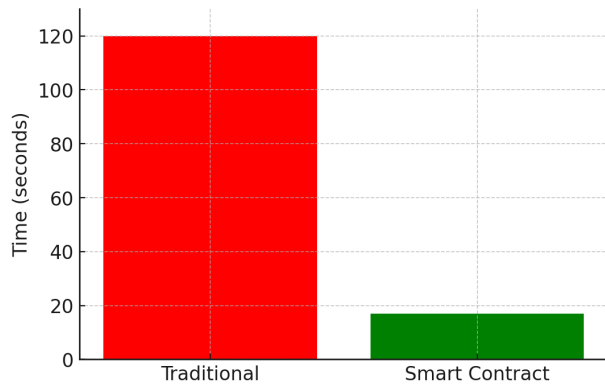


Fig. 3. Execution Time of Smart Contracts vs Traditional Transactions

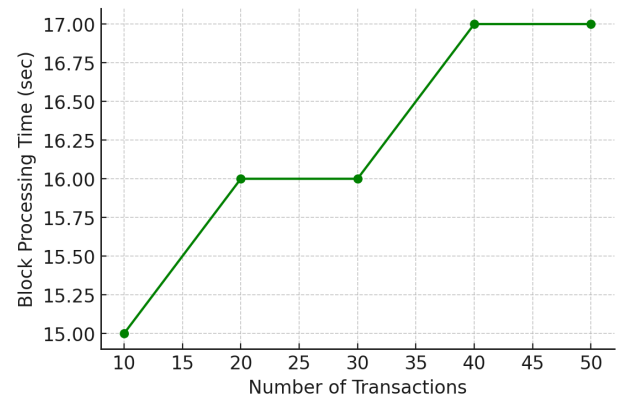


Fig. 5. Blockchain Network Performance with 50 Concurrent Transactions

Under network load, the system maintained a stable block processing time of 15–17 seconds even with 50 concurrent transactions. This consistency demonstrates the efficiency of the Proof of Authority (PoA) consensus, ensuring reliable scalability for real estate applications.

4. Lender & Inspector Approval Process

2. Gas Fees & Cost Efficiency

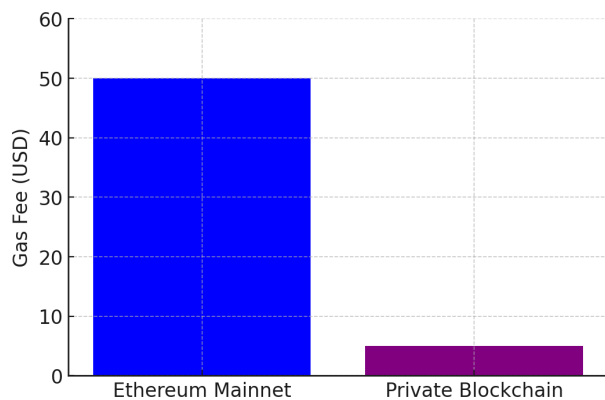


Fig. 4. Gas Fees for Property Transactions

A comparative analysis of transaction costs reveals a substantial reduction when shifting from the Ethereum Mainnet to a private blockchain environment. While the average gas fee on the public Ethereum network is approximately \$50 per transaction, the private blockchain alternative operates at only about \$5 per transaction reflecting an 85% decrease in cost. This reduction highlights the economic viability and sustainability of private blockchain adoption in large-scale real estate applications.

3. Scalability & Network Performance

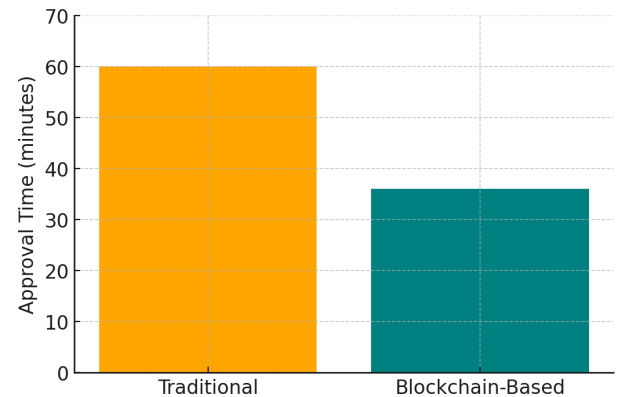


Fig. 6. Approval Time Comparison: Traditional vs Blockchain-Based

Property verification and approval time was reduced from about 60 minutes in the traditional process to nearly 36 minutes using the blockchain-based framework, representing an overall improvement of roughly 40%. This acceleration is achieved through a three-step approval mechanism involving the lender, inspector, and seller, which enhances trust, prevents fraudulent listings, and ensures regulatory compliance.

5. Tokenization & Ownership Transfer

Property tokenization and ownership verification using ERC-721 NFTs were completed in under five minutes, ensuring transparent, tamper-proof, and easily verifiable ownership records through blockchain-based documentation.

MetaMask integration achieved a 100% success rate in connecting Web3 wallets for property transactions, simplifying blockchain interaction for non-technical users and enhancing overall accessibility and adoption.

VII. FUTURE SCOPE AND CHALLENGES

A. Future Scope

- **Transforming Leasing and Rentals:** Blockchain can automate leasing through smart contracts, reducing broker dependency and transaction costs. Integration with decentralized MLS platforms and compliance systems can enhance transparency, prevent fraud, and standardize agreements.
- **Reforming Land Administration:** With nearly 70% of people lacking formal ownership records, blockchain offers immutable and tamper-proof registries. Future work will focus on Layer-2 solutions and optimized smart contracts for faster, cost-efficient, and transparent land transfers.

B. Challenges

- **Regulatory and Legal Barriers:** Lack of standard protocols and legal recognition for blockchain-based transactions hinders adoption. Translating property laws into smart contracts remains complex.
- **Industry Resistance:** High costs, limited awareness, and conservative practices slow implementation. Greater collaboration among legal, technical, and policy experts is needed.
- **Data Privacy:** Blockchain's immutability conflicts with data protection laws such as GDPR. Balancing transparency with privacy requires hybrid and off-chain storage models.

VIII. CONCLUSION

Blockchain is revolutionizing the real estate industry by addressing persistent issues such as inefficiencies, lack of transparency, and fraud. Its implementation guarantees secure, tamper-proof records, minimizing fraud risks and legal disputes in property transactions. Smart contracts automate key processes, reducing dependency on intermediaries and improving transaction efficiency. Tokenization introduces a new era of fractional ownership, increasing market liquidity and accessibility. The BlockRealty project leverages the Ethereum blockchain to introduce a three-step approval mechanism, significantly enhancing security and transparency in real estate dealings. However, a key limitation is the absence of fractional ownership support, which may prevent smaller investors from entering the market due to high capital requirements.

ACKNOWLEDGMENT

Special thanks to **Mr. Piyush Raj** for efficiently managing the student team and ensuring smooth coordination throughout the research work. His dedication and leadership were instrumental in the successful completion of this work.

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